AIML Applications and PEAS Analysis

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INTRODUCTION:

Artificial Intelligence (AI) is a branch of computer science that focuses on developing machines and software systems capable of performing tasks that normally require human intelligence. These tasks include perception, reasoning, learning, problem-solving, decision-making, planning and understanding information.

Machine Learning (ML) is a subfield of Artificial Intelligence in which machines learn patterns from data and improve their performance through experience rather than being explicitly programmed for every possible situation.

An intelligent agent is a system that perceives its surrounding environment through sensors and performs actions through actuators in order to achieve a particular objective.

The general working of an intelligent agent can be represented as:

Environment → Sensors → Intelligent Agent → Actuators → Environment The PEAS framework is an important method used to describe an intelligent agent.

PEAS stands for:

- **P – Performance Measure**
- **E – Environment**
- **A – Actuators**
- **S – Sensors**

Performance Measure

Performance Measure defines how the success of an intelligent agent is evaluated.

Environment

Environment represents the surroundings in which the agent operates.

Actuators

Actuators are the components through which the agent performs actions.

Sensors

Sensors allow the agent to perceive and collect information from its environment.

This report analyzes two important real-life applications of Artificial Intelligence and Machine Learning:

1. AI-Based Smart Museum & Heritage Preservation System:
2. AI-Based Smart Parking Management System

The analysis includes intelligent agents, environments, PEAS models, types of AI, types of intelligent agents, environmental characteristics, ethics, morality.

APPLICATION 1:

AI-Based Smart Museum & Heritage Preservation System:

Museums and heritage sites contain valuable objects, paintings, sculptures and historical structures that need continuous care. Artificial Intelligence can help museum staff monitor these places without checking everything manually.

An AI-based museum and heritage preservation system can use cameras, environmental sensors, drones and intelligent software to identify damage, control environmental conditions, manage visitors and improve security. The system can continuously collect information and alert staff whenever something unusual is detected.

The main aim of this system is to protect historical objects, reduce the chances of damage, improve visitor management and support conservation work.

1.1 Five Intelligent Agents:

Sr. No.	Intelligent Agent	Function
1.	Artifact Condition Monitoring Agent	Checks the overall condition of paintings, sculptures and other museum objects.
2.	Artifact Damage Detection Agent	Finds visible problems such as cracks, scratches, discoloration and deformation.
3.	Detects cracks, scratches, discoloration and other damage	Maintains suitable temperature, humidity and air conditions for artifact preservation.
4.	Visitor Flow Management Agent	Monitors visitor movement and helps control overcrowding in museum areas.
5.	Security & Intrusion Detection Agent	Identifies unauthorized entry or suspicious activity around protected museum areas.

1.2 Five Environments:

Sr. No.	Environment	Description
1.	Art Gallery	Indoor space where paintings, sculptures and other artworks are displayed for visitors.
2.	Historical Artifact Storage Room	Secure, controlled area where valuable and fragile historical artifacts are stored.
3.	Museum Exhibition Hall	Large indoor area where multiple exhibits are displayed and visitors move between them.
4.	Archaeological Heritage Site	Outdoor historical location containing ancient structures, monuments or archaeological remains.
5.	Museum Security & Control Room	Central facility where security systems, cameras and museum monitoring data are supervised.

1.3 10 Best PEAS combinations — Application 1

Combination 1

Artifact Condition Monitoring Agent & Art Gallery

Performance Measure (P):

- Accuracy of checking artifact condition
- Early identification of deterioration
- Percentage of gallery area monitored
- Number of false alerts
- Time taken to report a problem

Environment (E):

- Indoor art gallery
- Paintings and sculptures
- Artificial and natural lighting
- Visitors moving around the exhibits
- Controlled but changing humidity

Actuators (A):

- Send condition alerts to staff
- Show artifact status on the monitoring screen
- Request a detailed inspection
- Update the artifact record
- Notify the conservation team

Sensors (S):

- High-resolution camera
- Temperature sensor
- Humidity sensor
- Light sensor
- Vibration sensor

Description:
The agent regularly observes artworks and checks whether their condition has changed. If an artwork shows unusual discoloration, movement or environmental stress, the system informs the museum staff so that they can inspect it.

Combination 2

Artifact Condition Monitoring Agent & Archaeological Heritage Site

Performance Measure (P):

- Accuracy of condition assessment
- Area covered during monitoring
- Early identification of structural changes
- Detection response time
- Reliability of collected data

Environment (E):

- Outdoor historical site
- Ancient structures and monuments
- Changing weather conditions
- Tourist movement

- Dust and environmental pollution

Actuators (A):

- Highlight damaged locations on a map
- Send alerts to conservation staff
- Create inspection requests
- Store monitoring reports
- Schedule another inspection

Sensors (S):

- Drone camera
- LiDAR sensor
- Temperature sensor
- Humidity sensor
- GPS

Description:
The agent can use drone images and other sensors to inspect large heritage areas. It identifies places where cracks, weathering or other visible changes may have started and reports them to conservation staff.

Combination 3

Artifact Damage Detection Agent & Historical Artifact Storage Room

Performance Measure (P):

- Accuracy of damage detection
- Speed of inspection
- Detection of small cracks and scratches
- False alarm rate
- Number of artifacts checked

Environment (E):

- Secure storage room
- Fragile historical objects
- Storage shelves and cabinets

- Controlled temperature and humidity
- Very limited human movement

Actuators (A):

- Mark an artifact as damaged
- Inform the conservator
- Create an inspection request
- Update the artifact database
- Record the location of the affected object

Sensors (S):

- High-resolution camera
- 3D scanner
- Temperature sensor
- Humidity sensor
- RFID reader

Description:
The agent compares current images of stored artifacts with earlier records. If it notices a new crack, scratch or change in shape, it marks the artifact for further examination.

Combination 4

Artifact Damage Detection Agent & Art Gallery

Performance Measure (P):

- Damage detection accuracy
- Detection speed
- Number of false alarms
- Number of artworks monitored
- Quality of visual evidence

Environment (E):

- Public exhibition gallery
- Display cases and open displays
- Paintings and sculptures

- Artificial lighting
- Continuous visitor movement

Actuators (A):

- Generate a damage alert
- Show the affected artifact on the dashboard
- Inform gallery staff
- Start a closer camera inspection
- Save the related image

Sensors (S):

- CCTV camera
- High-resolution camera
- Motion sensor
- Vibration sensor
- Light sensor

Description:
The agent watches displayed objects and looks for sudden physical changes. For example, unusual vibration or a visible change in an artwork can trigger an alert for museum staff.

Combination 5

Climate Control Agent & Historical Artifact Storage Room

Performance Measure (P):

- Temperature stability
- Humidity control accuracy
- Energy consumption
- Artifact preservation quality
- Response time to environmental changes

Environment (E):

- Enclosed storage room
- Fragile historical objects
- Controlled surroundings

- Limited human activity
- Long-term artifact storage

Actuators (A):

- Adjust air-conditioning
- Start the humidifier
- Start the dehumidifier
- Control ventilation
- Send environmental warnings

Sensors (S):

- Temperature sensor
- Humidity sensor
- CO_2 sensor
- Air-quality sensor

Description:
The agent keeps the storage room within suitable environmental limits. When temperature or humidity moves outside the required range, it automatically adjusts the equipment or informs museum staff.

Combination 6

Climate Control Agent & Art Gallery

Performance Measure (P):

- Stability of temperature and humidity
- Protection of displayed artifacts
- Energy efficiency
- Speed of environmental correction
- Number of abnormal conditions

Environment (E):

- Exhibition rooms
- Large number of visitors
- Artificial lighting

- Different types of artifacts
- Changing indoor conditions

Actuators (A):

- Adjust HVAC settings
- Control ventilation
- Operate humidifiers or dehumidifiers
- Send warnings to staff
- Record environmental changes

Sensors (S):

- Temperature sensor
- Humidity sensor
- CO_2 sensor
- Light sensor
- Air-quality sensor

Description:
The agent observes the gallery environment and makes small adjustments whenever conditions become unsuitable for the displayed artifacts. This helps maintain a stable environment without continuous manual checking.

Combination 7

Visitor Flow Management Agent & Museum Exhibition Hall

Performance Measure (P):

- Accuracy of visitor counting
- Accuracy of crowd detection
- Reduction in overcrowding
- Waiting time
- Speed of response

Environment (E):

- Large exhibition hall
- Multiple displays

- Moving visitors
- Entry and exit points
- Areas with different visitor densities

Actuators (A):

- Display crowd warnings
- Suggest another route
- Control entry gates
- Inform museum staff
- Update the visitor monitoring dashboard

Sensors (S):

- CCTV cameras
- People-counting camera
- Infrared sensor
- Door sensor
- Occupancy data

Description:
The agent keeps track of how many people are present in different parts of the exhibition hall. If an area becomes too crowded, it can suggest another route or alert staff to control entry.

Combination 8

Visitor Flow Management Agent & Archaeological Heritage Site

Performance Measure (P):

- Accuracy of crowd estimation
- Protection of restricted areas
- Visitor safety
- Waiting time
- Coverage of important areas

Environment (E):

- Outdoor heritage location
- Tourist groups

- Restricted zones
- Multiple walking paths
- Changing weather

Actuators (A):

- Display recommended routes
- Send crowd alerts
- Control access gates
- Notify site authorities
- Suggest less crowded areas

Sensors (S):

- CCTV camera
- People-counting camera
- GPS
- Motion sensor
- Access-control sensor

Description:
The agent observes tourist movement around the heritage site and helps prevent excessive crowding. It can also identify when visitors move toward restricted or sensitive areas.

Combination 9

Security and Intrusion Detection Agent & Museum Security and Control Room

Performance Measure (P):

- Intrusion detection accuracy
- False alarm rate
- Response time
- Number of areas monitored
- Number of security incidents detected

Environment (E):

- Museum control room
- Multiple galleries

- Restricted rooms
- Entry and exit points
- After-hours operation

Actuators (A):

- Trigger a security alarm
- Lock restricted doors
- Inform security personnel
- Display the relevant camera feed
- Store incident information

Sensors (S):

- CCTV cameras
- Motion sensors
- Door sensors
- RFID sensors
- Access-control readers

Description:
The agent combines information from cameras and access systems to identify suspicious activity. If an unauthorized person enters a restricted area, security staff receive an immediate alert.

Combination 10

Security and Intrusion Detection Agent & Historical Artifact Storage Room

Performance Measure (P):

- Accuracy of unauthorized-access detection
- Response time
- False alarm rate
- Protection of valuable objects
- Reliability of access records

Environment (E):

- Restricted storage room

- Valuable historical artifacts
- Authorized staff only
- Controlled access points
- Low human activity

Actuators (A):

- Lock the door
- Activate an alarm
- Notify security personnel
- Start or increase recording
- Save the access event

Sensors (S):

- RFID reader
- Door sensor
- Motion sensor
- CCTV camera
- Access-control or biometric sensor

Description:
The agent checks who enters the storage area and monitors movement inside it. An unauthorized access attempt can automatically activate security measures and inform the responsible staff.

1.4 Societal Impact:AI-Based Smart Museum & Heritage Preservation System

Positive Impact

1. Protects cultural heritage: AI can help detect damage early and support the preservation of valuable artifacts and historical sites.
2. Improves visitor experience: Monitoring visitor movement can reduce overcrowding and make museum visits more comfortable.
3. Improves security: The system can identify suspicious activity and unauthorized entry in protected areas.

4. Reduces manual work: Staff do not have to continuously monitor every artifact, room or security camera manually.
5. Faster response: Alerts can help staff notice environmental problems or possible damage before they become serious.

Negative Impact

1. Privacy concerns: Cameras used for visitor monitoring may make people uncomfortable if their movements are recorded.
2. High cost: Cameras, sensors, drones and other equipment can be expensive to install and maintain.
3. False alerts: The system may sometimes mistake normal changes for artifact damage or unusual activity.
4. Dependence on technology: A sensor, camera or network failure can affect monitoring and may cause important information to be missed.

1.5 PEAS SUMMARY – AI-BASED SMART MUSEUM & HERITAGE PRESERVATION SYSTEM:

Sr. No.	Agent	Environment	Main Sensors	Main Actuators
1.	Artifact Condition Monitoring Agent	Art Gallery	Camera, temperature, humidity, light, vibration sensors	Condition alerts, status display, inspection request
2.	Artifact Condition Monitoring Agent	Archaeological Heritage Site	Drone camera, LiDAR, temperature, humidity, GPS	Damage map, alerts, inspection request
3.	Artifact Damage Detection Agent	Historical Artifact Storage Room	Camera, 3D scanner, temperature, humidity, RFID	Damage alert, inspection request, database update
4.	Artifact Damage Detection Agent	Art Gallery	CCTV, camera, motion, vibration, light sensors	Damage alert, dashboard update, camera inspection

5.	Climate Control Agent	Historical Artifact Storage Room	Temperature, humidity, CO_2, air-quality sensors	AC control, humidifier/dehumidifier, ventilation
6.	Climate Control Agent	Art Gallery	Temperature, humidity, CO_2, light, air-quality sensors	HVAC control, ventilation, environmental alerts
7.	Visitor Flow Management Agent	Museum Exhibition Hall	CCTV, people-counting, infrared, door, occupancy sensors	Crowd warnings, route suggestions, entry control
8.	Visitor Flow Management Agent	Archaeological Heritage Site	CCTV, people-counting, GPS, motion, access sensors	Crowd alerts, route display, access control
9.	Security & Intrusion Detection Agent	Museum Security & Control Room	CCTV, motion, door, RFID, access-control sensors	Security alarm, door lock, security notification
10.	Security & Intrusion Detection Agent	Historical Artifact Storage Room	RFID, door, motion, CCTV, access-control sensors	Door lock, alarm, security notification

1.5 ETHICS AND MORALITY – AI-BASED SMART MUSEUM & HERITAGE PRESERVATION SYSTEM

1. Safety

The system should help protect valuable artifacts, museum visitors and staff. Alerts should be reliable so that important problems are not missed.

2. Accountability

There should be a clear person or team responsible for checking important alerts and taking action when the system detects a problem.

3. Privacy

Cameras used for visitor monitoring should not collect or store unnecessary personal information. Visitor data should be protected from unauthorized access.

4. Bias

The monitoring system should work properly in different lighting conditions, crowd sizes and locations so that it does not perform poorly in certain areas.

5. Transparency

Museum staff should understand what the system is monitoring and why an alert has been generated. The system should not make its decisions completely unclear to the staff.

6. Human Control

Important decisions, such as moving an artifact, closing an area or responding to a security alert, should be checked by museum staff instead of being handled entirely by the AI system.

7. Fairness

The system should treat visitors fairly and should not unnecessarily target particular people or groups during visitor or security monitoring.

APPLICATION 2:

AI-Based Smart Parking Management System

2. Introduction

Finding a parking space can be difficult, especially in places such as shopping malls, offices and railway stations. Drivers may have to spend a lot of time looking for an empty space, which can also increase traffic inside the parking area.

An AI-based smart parking system can make this process easier. It can use cameras and sensors to check parking spaces, guide drivers, identify vehicles parked in the wrong places and manage vehicles at entry and exit gates. It can also use previous parking information to estimate how busy a parking area will be.

The main purpose of this system is to make parking more organized, reduce the time spent searching for a space and help parking staff manage the area more easily.

2.1 Five Intelligent Agents:

Sr. No.	Intelligent Agent	Function
1.	Parking Space Detection Agent	Checks which parking spaces are free and which ones are already occupied.
2.	Parking Guidance Agent	Helps drivers reach a suitable empty parking space.
3.	Parking Violation Detection Agent	Finds vehicles that are parked in restricted or incorrect areas.
4.	Entry and Exit Management Agent	Keeps track of vehicles entering and leaving the parking facility.
5.	Parking Availability Prediction Agent	Uses current and previous parking data to estimate future parking availability.

2.2 Five Environments

Sr. No.	Environment	Description
1.	Shopping Mall Parking	Parking area used by people who come to a shopping mall.
2.	Railway Station Parking	Parking area where passengers leave their vehicles while travelling by train.
3.	Office Parking	Parking space mainly used by employees and people visiting an office.
4.	Hospital Parking	Parking facility used by patients, visitors, doctors and hospital staff.
5.	Public Street Parking	Marked roadside parking spaces available for public use.

2.3 PEAS ANALYSIS – 10 Selected Combinations

Combination 1

Parking Space Detection Agent & Shopping Mall Parking

Performance Measure (P):

- Correct identification of free and occupied spaces
- Quick detection of changes
- Number of parking spaces monitored
- Few incorrect results

Environment (E):

- Shopping mall parking area
- Many parking spaces
- Cars entering and leaving
- Different lighting conditions

Actuators (A):

- Show available spaces
- Update the parking display
- Send information to parking staff

- Update the parking map

Sensors (S):

- CCTV camera
- Ultrasonic sensor
- Vehicle detection sensor
- Parking-space sensor

Description:
The agent checks the parking spaces and tells the system which spaces are empty and which are occupied. This information can then be shown to drivers so they do not have to search unnecessarily.

Combination 2

Parking Space Detection Agent & Hospital Parking

Performance Measure (P):

- Accuracy of parking-space detection
- Speed of detecting changes
- Number of spaces monitored
- Correct availability information

Environment (E):

- Hospital parking area
- Cars and ambulances
- Reserved parking spaces
- Vehicles entering and leaving regularly

Actuators (A):

- Update parking displays
- Show free spaces
- Inform parking staff
- Mark reserved spaces

Sensors (S):

- CCTV camera
- Ultrasonic sensor
- Vehicle detection sensor
- RFID sensor

Description:
The agent keeps track of the available spaces in a hospital parking area. It can also help staff keep reserved spaces, such as ambulance spaces, available when they are needed.

Combination 3

Parking Guidance Agent & Shopping Mall Parking

Performance Measure (P):

- Accuracy of directions
- Reduction in parking search time
- Efficient movement of vehicles
- Number of drivers successfully guided

Environment (E):

- Multi-level shopping mall parking
- Several parking sections
- Vehicles moving between sections
- Multiple entry and exit points

Actuators (A):

- Display directions
- Show available parking spaces
- Change digital signs
- Suggest a nearby empty space

Sensors (S):

- Parking-space sensors
- CCTV cameras

- Entry sensors
- Parking map

Description:
The agent uses information about available spaces to guide drivers toward a suitable parking spot. This can save time, especially in large multi-level parking areas.

Combination 4

Parking Guidance Agent & Railway Station Parking

Performance Measure (P):

- Correctness of parking directions
- Reduction in time spent searching
- Smooth movement of vehicles
- Number of vehicles guided

Environment (E):

- Railway station parking area
- High vehicle movement
- Different parking sections
- Busy periods during train timings

Actuators (A):

- Display parking directions
- Change electronic signs
- Suggest available spaces
- Warn about crowded sections

Sensors (S):

- CCTV cameras
- Parking sensors
- Entry and exit sensors
- Vehicle counters

Description:
The agent helps passengers find parking without driving around the entire parking area. It can direct them toward sections where spaces are currently available.

Combination 5

Parking Violation Detection Agent & Public Street Parking

Performance Measure (P):

- Accuracy of detecting violations
- Quick identification of wrongly parked vehicles
- Low false alarm rate
- Number of violations detected

Environment (E):

- Roadside parking spaces
- Moving traffic
- Restricted parking areas
- Different weather and lighting conditions

Actuators (A):

- Generate a violation alert
- Save vehicle evidence
- Inform parking authorities
- Record the location of the violation

Sensors (S):

- CCTV camera
- Number-plate camera
- Parking sensor
- Location data

Description:
The agent checks whether vehicles are parked according to the parking rules. If a vehicle is found in a restricted area, the system can record it and notify the concerned authority.

Combination 6

Parking Violation Detection Agent & Hospital Parking

Performance Measure (P):

- Accuracy of violation detection
- Speed of reporting
- Number of violations detected
- Low false alarm rate

Environment (E):

- Hospital parking area
- Reserved parking spaces
- Ambulance zones
- Staff and visitor parking

Actuators (A):

- Send alerts to security staff
- Capture vehicle images
- Record parking violations
- Notify the responsible person

Sensors (S):

- CCTV camera
- Number-plate camera
- Parking sensor
- RFID reader

Description:
The agent checks whether vehicles are using hospital parking spaces correctly. For example, it can detect when an unauthorized vehicle is parked in an ambulance or reserved space.

Combination 7

Entry and Exit Management Agent & Office Parking

Performance Measure (P):

- Fast processing of vehicles
- Correct vehicle identification
- Short waiting time
- Correct access decisions

Environment (E):

- Office parking facility
- Employees and visitors
- Entry and exit gates
- Busy office hours

Actuators (A):

- Open and close gates
- Allow or deny entry
- Generate parking tickets
- Update vehicle records

Sensors (S):

- RFID reader
- Number-plate camera
- Vehicle detection sensor
- Gate sensor

Description:
The agent manages vehicles at the office parking entrance and exit. It checks vehicle information and allows authorized vehicles to enter while keeping a record of parking activity.

Combination 8

Entry and Exit Management Agent & Railway Station Parking

Performance Measure (P):

- Speed of gate operation
- Correct vehicle identification
- Reduction in waiting queues

- Number of vehicles processed

Environment (E):

- Railway station parking
- Large number of vehicles
- Entry and exit gates
- Busy travel periods

Actuators (A):

- Open and close gates
- Issue parking tickets
- Update parking records
- Display parking instructions

Sensors (S):

- RFID reader
- Number-plate camera
- Vehicle sensor
- Gate sensor

Description:
The agent controls the movement of vehicles through the parking gates. It is especially useful during busy periods because it can reduce unnecessary waiting at the entrance and exit.

Combination 9

Parking Availability Prediction Agent & Shopping Mall Parking

Performance Measure (P):

- Accuracy of parking predictions
- Correct estimation of available spaces
- Reduction in search time
- Speed of prediction

Environment (E):

- Shopping mall parking
- Different shopping hours

- Weekdays and weekends
- Different numbers of arriving vehicles

Actuators (A):

- Display expected parking availability
- Send parking information
- Suggest less crowded sections
- Update the parking dashboard

Sensors (S):

- Previous parking records
- Parking-space sensors
- Entry and exit counters
- CCTV cameras

Description:
The agent looks at previous parking patterns and current vehicle activity to estimate how many spaces may be available at a particular time.

Combination 10

Parking Availability Prediction Agent & Office Parking

Performance Measure (P):

- Accuracy of predictions
- Accuracy during busy hours
- Reduction in parking delays
- Reliability of the predicted information

Environment (E):

- Office parking facility
- Fixed working hours
- Employee vehicles
- Visitor vehicles
- Morning and evening rush periods

Actuators (A):

- Display expected availability
- Send parking alerts
- Update the parking dashboard
- Suggest another parking area

Sensors (S):

- Previous parking records
- Entry and exit sensors
- Parking-space sensors
- Vehicle counters

Description:
The agent studies previous parking patterns along with current data to predict busy periods. This information can help employees know when parking is likely to be difficult.

2.4 Societal Impact:AI-Based Smart Parking Management System

Positive Impact

1. Saves drivers' time: The system can help drivers find available parking spaces without searching for a long time.
2. Reduces traffic inside parking areas: Better guidance can reduce unnecessary vehicle movement and congestion.
3. Improves parking management: Parking staff can easily monitor available spaces, vehicle entry and exit, and parking violations.
4. Improves safety: Proper parking guidance and organized vehicle movement can reduce confusion and the chances of minor accidents.
5. Better use of parking space: The system can provide information about unused spaces and help distribute vehicles more efficiently.

Negative Impact

1. Privacy concerns: Cameras and number-plate recognition may collect information about vehicles and their movements.
2. Technical failures: Incorrect sensor readings can show a space as free when it is actually occupied or vice versa.

3. False parking violations: The system may wrongly identify a vehicle as being parked incorrectly.
4. Implementation cost: Installing cameras, sensors, RFID systems and automated gates can require significant investment.
5. Dependence on the system: If the AI or network stops working, drivers and parking staff may have to manage the parking area manually.

2.5 PEAS SUMMARY – AI-BASED SMART PARKING MANAGEMENT SYSTEM:

Sr. No.	Agent	Environment	Main Sensors	Main Actuators
1.	Parking Space Detection Agent	Shopping Mall Parking	CCTV, ultrasonic, vehicle, parking-space sensors	Availability display, parking-map update
2.	Parking Space Detection Agent	Hospital Parking	CCTV, ultrasonic, vehicle, RFID sensors	Parking display, free-space update, staff alert
3.	Parking Guidance Agent	Shopping Mall Parking	Parking-space sensors, CCTV, entry sensors, parking map	Directions, digital signs, space suggestion
4.	Parking Guidance Agent	Railway Station Parking	CCTV, parking, entry/exit sensors, vehicle counters	Directions, electronic signs, availability display
5.	Parking Violation Detection Agent	Public Street Parking	CCTV, number-plate camera, parking sensor, location data	Violation alert, evidence recording, authority notification
6.	Parking Violation Detection Agent	Hospital Parking	CCTV, number-plate camera, parking sensor, RFID	Security alert, image capture, violation record
7.	Entry and Exit Management Agent	Office Parking	RFID, number-plate camera, vehicle, gate sensors	Gate control, ticket generation, vehicle records
8.	Entry and Exit Management Agent	Railway Station Parking	RFID, number-plate camera, vehicle, gate sensors	Gate control, ticket issuing, parking instructions

9.	Parking Availability Prediction Agent	Shopping Mall Parking	Parking records, parking-space sensors, entry/exit counters, CCTV	Availability display, parking alerts, dashboard update
10.	Parking Availability Prediction Agent	Office Parking	Parking records, entry/exit sensors, parking-space sensors, vehicle counters	Availability display, parking alerts, alternative parking suggestion

ETHICS AND MORALITY – AI-BASED SMART PARKING MANAGEMENT SYSTEM

1. Safety

The system should help maintain safe movement of vehicles and pedestrians. Parking alerts and gate controls should work correctly to avoid creating dangerous situations.

2. Accountability

There should be a responsible parking authority or staff member who can review system decisions, especially when a vehicle is marked as violating parking rules.

3. Privacy

Cameras and number-plate recognition can collect information about vehicles and people. This data should only be used when necessary and should be properly protected.

4. Bias

The system should be tested with different types of vehicles, parking conditions, lighting and weather so that it does not regularly make mistakes in certain situations.

5. Transparency

Drivers should be able to understand basic parking information, such as why a space is unavailable or why a parking violation has been detected.

6. Human Control

Parking staff should be able to review important alerts and override the system when necessary. The AI should assist staff rather than completely replace human judgment.

7. Fairness

The system should apply the same parking rules to all vehicles. It should not give unfair advantages or disadvantages to particular drivers without a valid reason.

LEGAL ISSUES

1. Privacy and data protection: Visitor images, vehicle details and other collected information should be properly protected and used only for valid purposes.
2. Camera surveillance: People should be informed when CCTV or other monitoring systems are being used.
3. Data security: Stored images, vehicle records and sensor data should be accessible only to authorized people.
4. Responsibility for AI decisions: If the system makes a wrong decision, such as falsely detecting artifact damage or a parking violation, there should be a clear process for human review.
5. Cultural heritage rules: The museum system should follow applicable rules when monitoring or working with protected historical sites and artifacts.
6. Parking regulations: The smart parking system should follow the parking laws and rules of the particular location.

COMPARISON OF BOTH AI APPLICATIONS

Parameter	**AI-Based Smart Museum & Heritage Preservation**	**AI-Based Smart Parking Management**
Main Objective	Protect historical artifacts, manage visitors and improve museum security	Find parking spaces easily and improve parking management
Main AI Techniques	Computer vision, sensor data analysis, pattern detection	Computer vision, sensor data analysis, prediction
Number of Agents	5	5
Number of Environments	5	5
PEAS Combinations	10	10
Main Sensors	Cameras, temperature and humidity sensors, LiDAR, motion sensors	CCTV cameras, parking sensors, RFID, number-plate cameras
Main Actuators	Alerts, HVAC control, access control, dashboard updates	Parking displays, gate control, alerts, directions
Major Benefit	Better protection of artifacts and smoother museum management	Less time spent searching for parking and better traffic flow

Major Risk	Incorrect damage detection or environmental readings may affect conservation decisions	Incorrect parking detection may result in false alerts or wrong actions
Privacy Concern	Visitor images and surveillance data may raise privacy concerns	Vehicle images, number plates and parking records may raise privacy concerns
Bias Concern	System may perform differently under different lighting, locations or crowd conditions	System may make more errors with different vehicles, lighting or weather conditions
Major Legal Issue	Privacy, surveillance and cultural heritage protection rules	Privacy, surveillance and parking regulations
Human Supervision	Required before major conservation, security or artifact-related decisions	Required before major parking enforcement or disputed violation decisions

OVERALL CONCLUSION

The PEAS analysis shows how Artificial Intelligence can be applied to different real-life situations by combining intelligent agents with suitable environments, sensors and actuators. In this report, two applications were studied: the AI-Based Smart Museum & Heritage Preservation System and the AI-Based Smart Parking Management System.

In the museum application, different agents can help monitor artifacts, detect possible damage, maintain suitable environmental conditions, manage visitor movement and improve security. The PEAS combinations show how the same type of agent can perform different tasks depending on the environment in which it operates.

In the parking application, AI agents can detect available spaces, guide drivers, identify parking violations, manage entry and exit, and predict future parking availability. These functions can make parking more organized and reduce the time drivers spend searching for spaces.

From this analysis, it is clear that the effectiveness of an intelligent agent depends not only on the agent itself but also on the environment, sensors, actuators and performance measures used with it. At the same time, AI systems are not completely independent or error-free. Sensor failures, incorrect predictions, privacy concerns and unusual situations can affect their performance. Therefore, human supervision is still important.

Overall, the PEAS framework provides a simple way to understand how an AI system observes its surroundings, makes decisions and performs actions to achieve its goal. Both applications show that AI can be useful in practical problems when it is designed according to the needs and limitations of the environment.